Data Visualization

Data Visualization: is the graphical representation of data, by using of common graphics

Data visualization is a crucial part of data analysis, as it allows us to better understand and communicate patterns, trends, and insights in our data.

Types of Plots

Function & Description

- Bar ==> Make a bar plot.
- Barh ==> Make a horizontal bar plot.
- Boxplot ==> Make a box and whisker plot.
- Hist ==> Plot a histogram.
- hist2d ==> Make a 2D histogram plot.
- Pie ==> Plot a pie chart.
- Plot ==> Plot lines and/or markers to the Axes.
- Scatter ==> Make a scatter plot of x vs y.
- Stackplot ==> Draws a stacked area plot.

Visualization with Matplotlib

Matplotlib

Matplotlib is a low level graph plotting library in python that serves as a visualization utility.

Matplotlib is open source and we can use it freely.

Libraries required

- pip install matplotlib

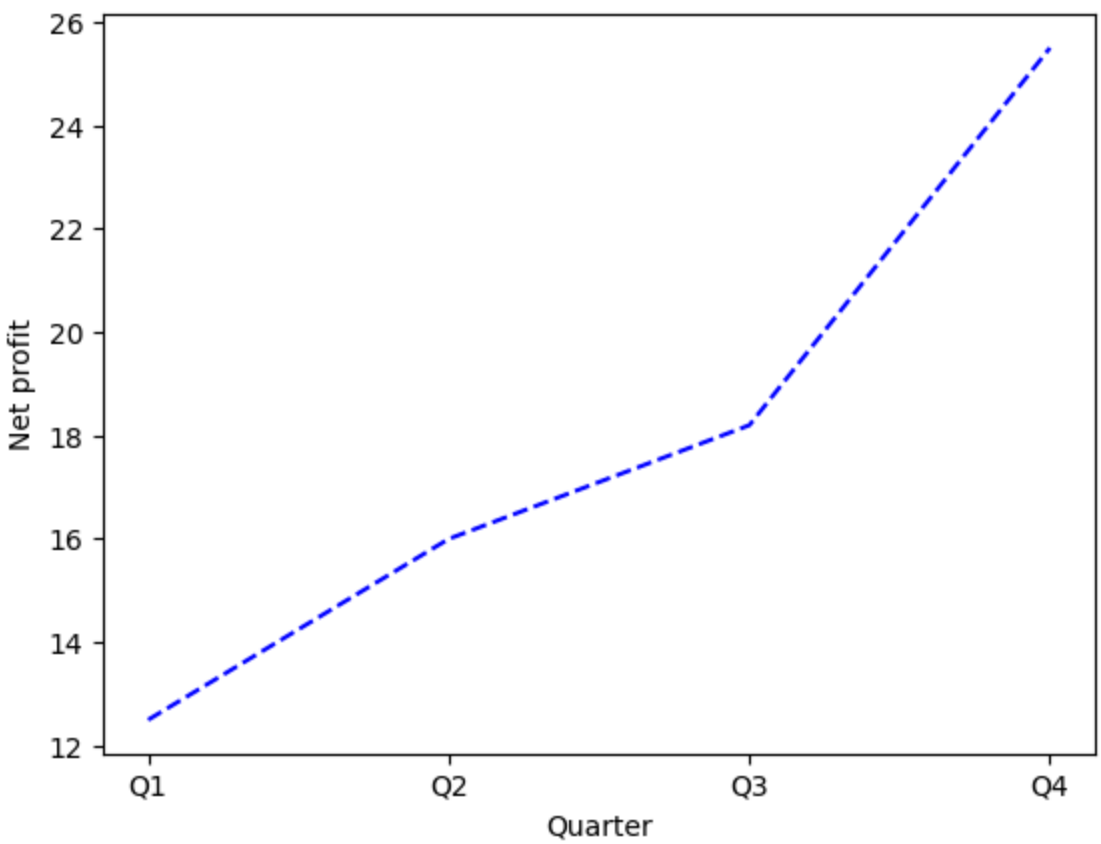
Importing Libraries

In [ ]:
```
# importing libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

Line Plot

Line charts can be used to show trends within a continuous data set and can be applied to a wide variety of categories, including daily number of visitors to a site or variations in stock prices.

In [ ]:
```
plt.plot(['Q1', 'Q2', 'Q3', 'Q4'],
         [12.5, 16.0, 18.2, 25.5],'b--')

plt.xlabel("Quarter")
plt.ylabel("Net profit")
plt.show()
```

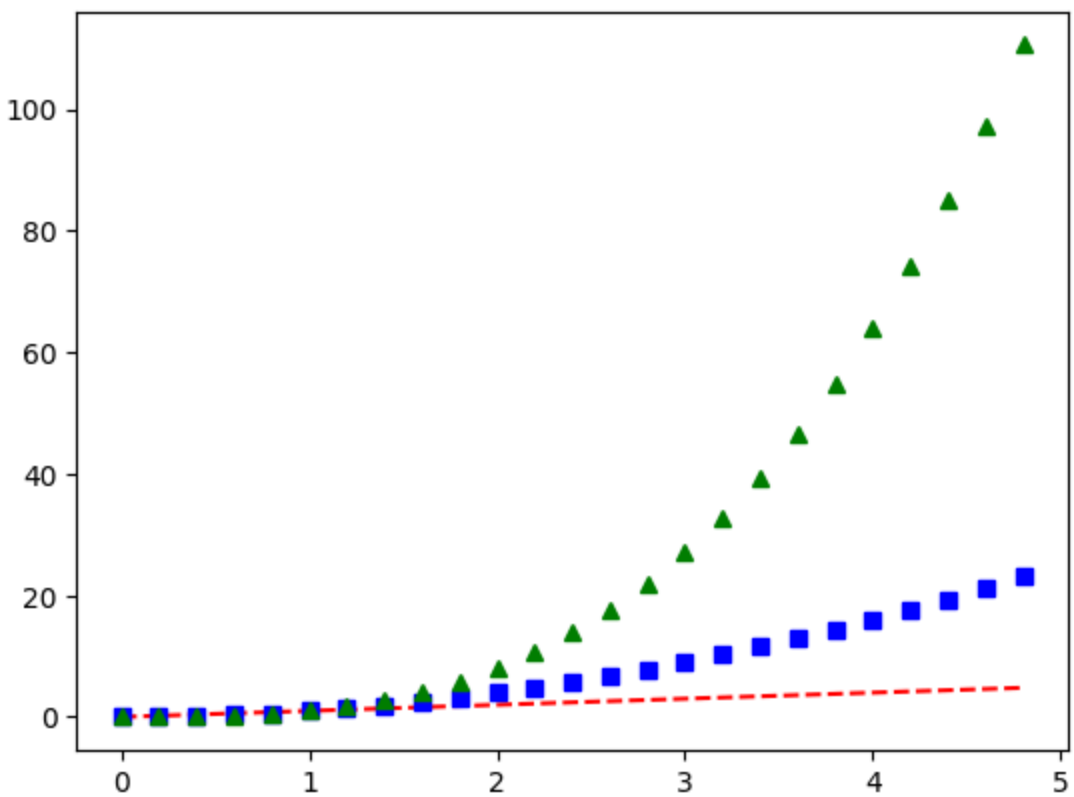


Formatting the style of your plot

To plot symbols rather than lines, provide an additional string argument.

- symbols (- , –, -., , . , , , o , ^ , v , < , > , s , + , x , D , d , 1 , 2 , 3 , 4 , h , H , p , | , _)
- colors (b, g, r, c, m, y, k, w)

In [ ]:
```
# evenly sampled time at 200ms intervals
t = np.arange(0., 5., 0.2)

# red dashes, blue squares and green triangles
plt.plot(t, t, 'r--')
plt.plot(t, t**2, 'bs')
plt.plot(t, t**3, 'g^')

plt.show()
```



Bar Plot

-A bar plot or bar chart uses rectangular bars to visualize data.

- It helps to get an easy comparison between different categories
- Bar charts can be displayed vertically or horizontally.

Vertical Bar Chart (Column Chart): **bar()**

The categories are typically on the horizontal (x) axis.

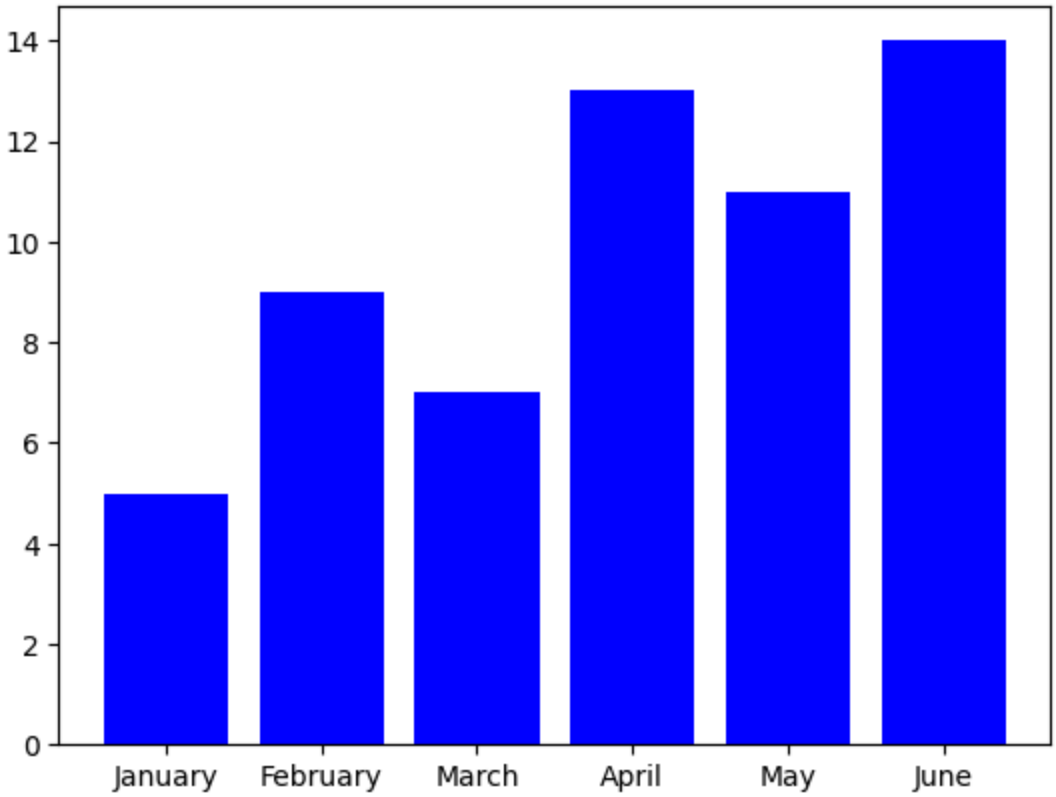
The values are on the vertical (y) axis.

Horizontal Bar Chart: (better for long category names) **barh()**

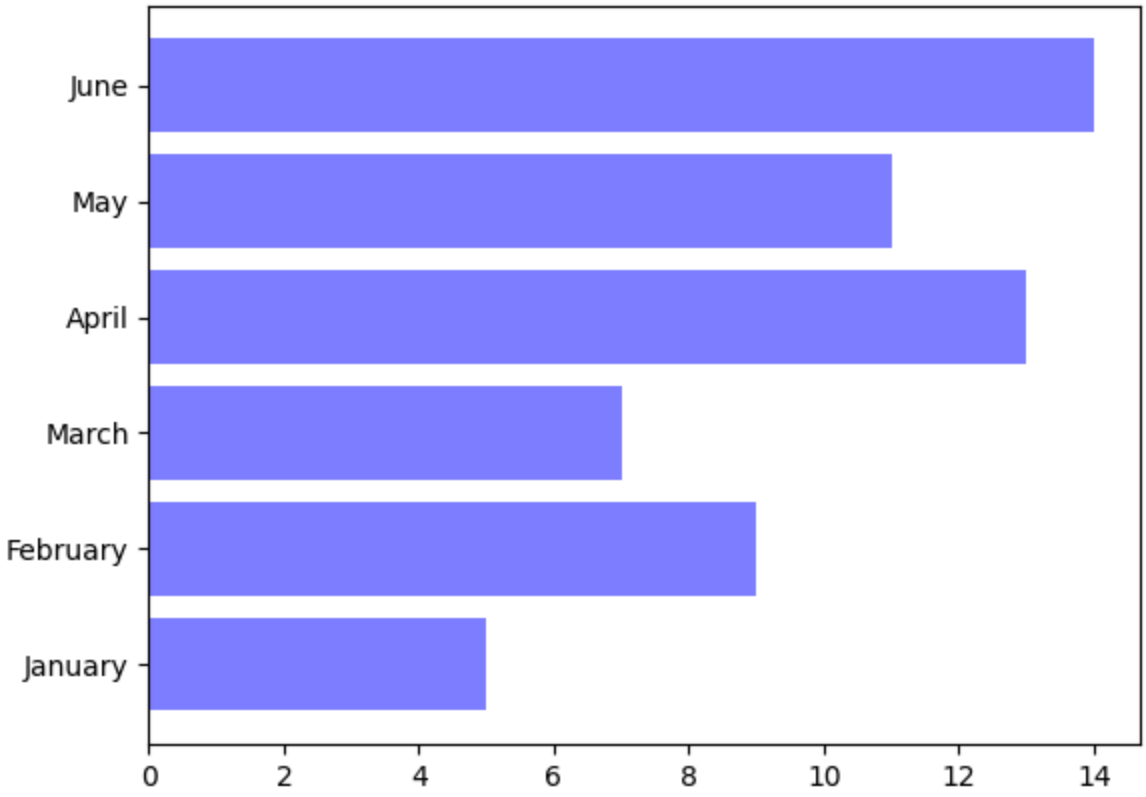
The categories are on the vertical (y) axis.

The values are on the horizontal (x) axis.

In [ ]:
```
# 1. Define the data
months = ['January', 'February', 'March', 'April', 'May', 'June']
# Website views (in thousands)
website_views = [5, 9, 7, 13, 11, 14]
bars = plt.bar(months, website_views, color='blue', alpha=1)
```



In [ ]:
```
# Horizontal Bar Chart
months = ['January', 'February', 'March', 'April', 'May', 'June']
# Website views (in thousands)
website_views = [5, 9, 7, 13, 11, 14]
bars = plt.barh(months, website_views, color='blue', alpha=0.5)
```

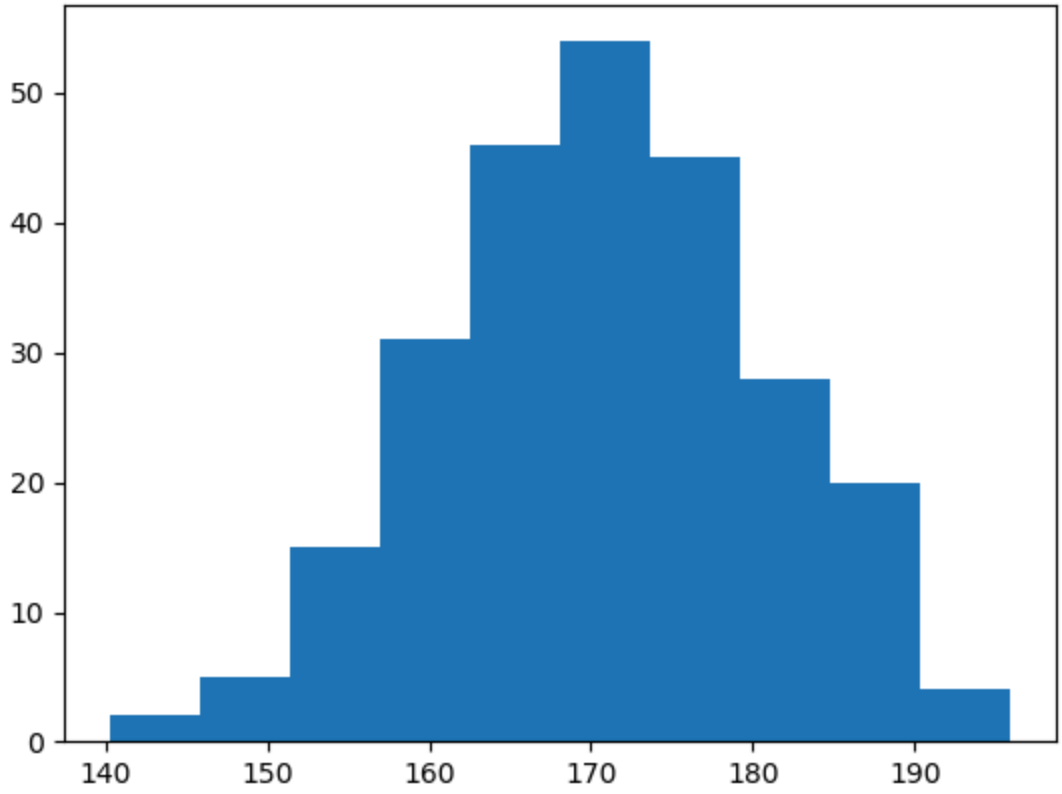


Histogram

Show the frequency distributions of a single continuous variable

you can use the hist() function to create histograms

Example: Say you ask for the height frequency of 250 people



Scatter Plot

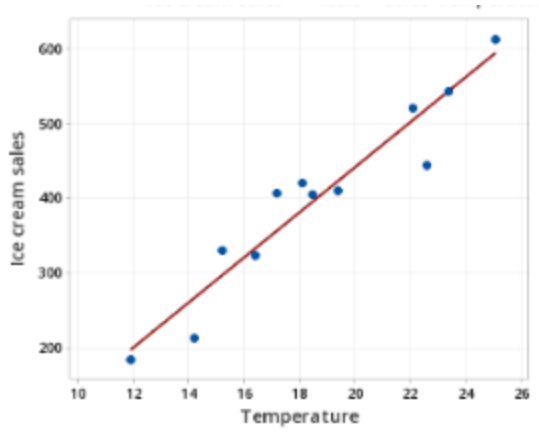
is one the most common plots, it used to display the relationship between two numerical variables.

One of the variables is represented on x-axis and another on y-axis.

you can use the scatter() function to draw a scatter plot.

Example: here is the visualization of the Ice Cream Sales vs. Temperature data

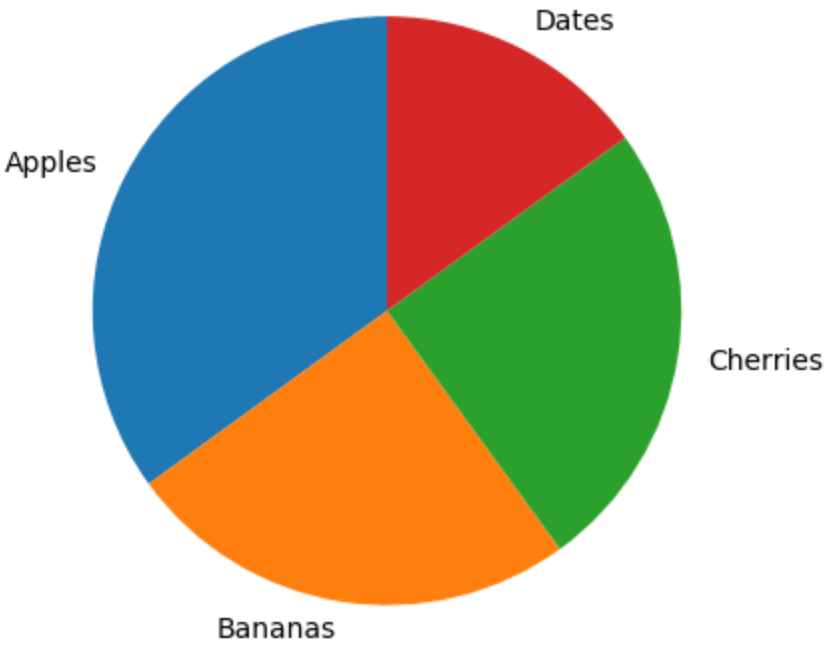
The visualization demonstrates a strong positive correlation. This means that as the temperature increases, the ice cream sales reliably increase as well.



pie chart

is a circular graphical view of data, by dividing into different segments according to the ratio of data provided. The total value of the pie is 100 and the segments tell the fraction of the whole pie.

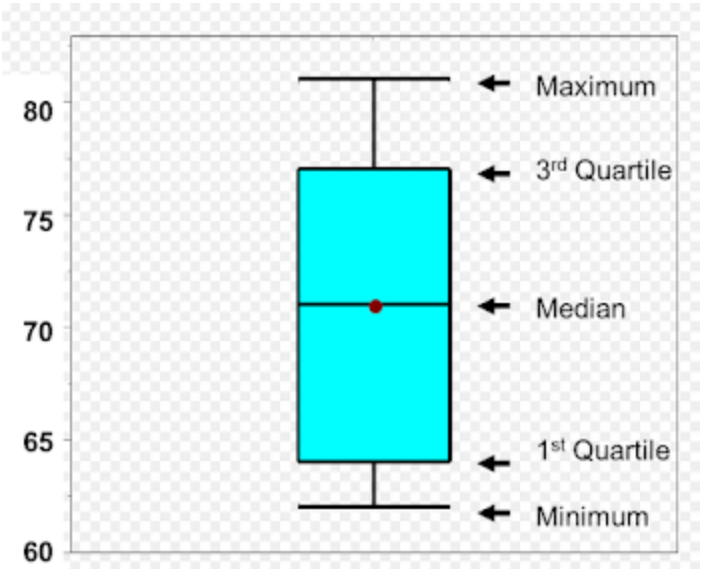
you can use the pie() function to draw pie charts



Box Plot

Box plot shows how the data is distributed, displays the five-number summary of a set of data: minimum, first quartile, median, third quartile, and maximum.

We can use it to identify outliers



Let’s visualize employees data

We have a csv file containing data about employees' salary

Read-in Data

In [ ]:
```
df= pd.read_csv("employees.csv")
df.head()
```

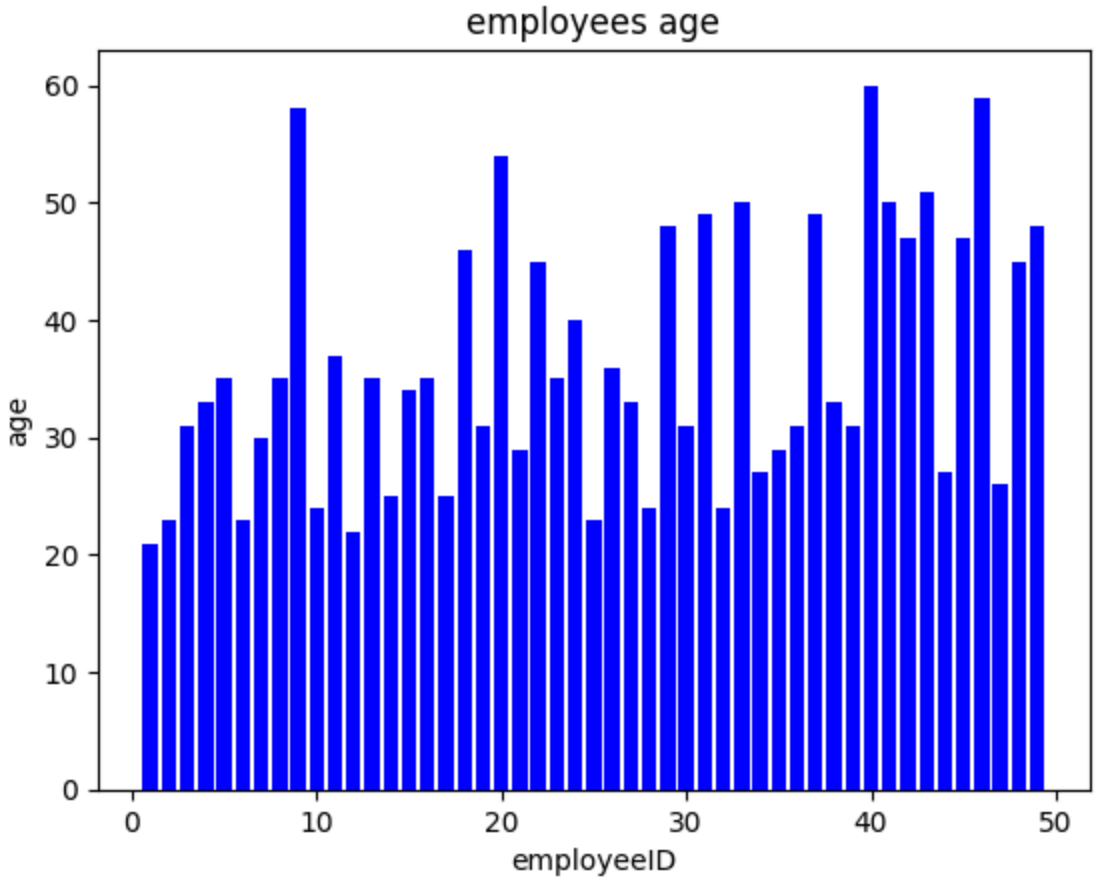
Out[ ]:

	employeeID	gender	age	salary
0	1	Male	21	2
1	2	Male	23	3
2	3	Male	31	5
3	4	Male	33	5
4	5	Male	35	6

In [ ]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 49 entries, 0 to 48
Data columns (total 4 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   employeeID  49 non-null     int64 
 1   gender      49 non-null     object
 2   age         49 non-null     int64 
 3   salary      49 non-null     int64 
dtypes: int64(3), object(1)
memory usage: 1.7+ KB
```

In [ ]:
```
# Display Age values
plt.bar(df['employeeID'], df['age'], color='blue')
# set title
plt.title("employees age")
# set labels
plt.xlabel("employeeID")
plt.ylabel("age")
#plt.ylabel("Age", fontsize=14, color="blue")
# show the plot
plt.show()
```

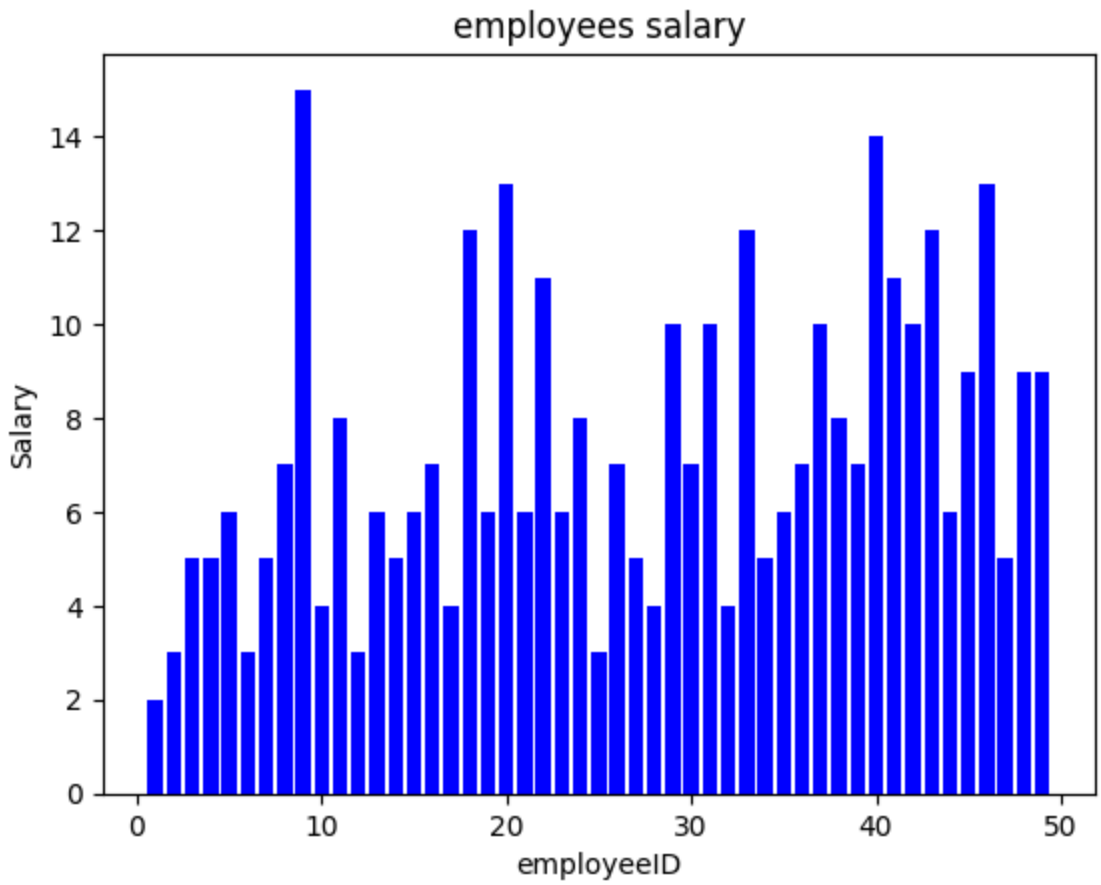


The minimum age is >20

The max age is 60

We can say that the employees age are between 20 to 60

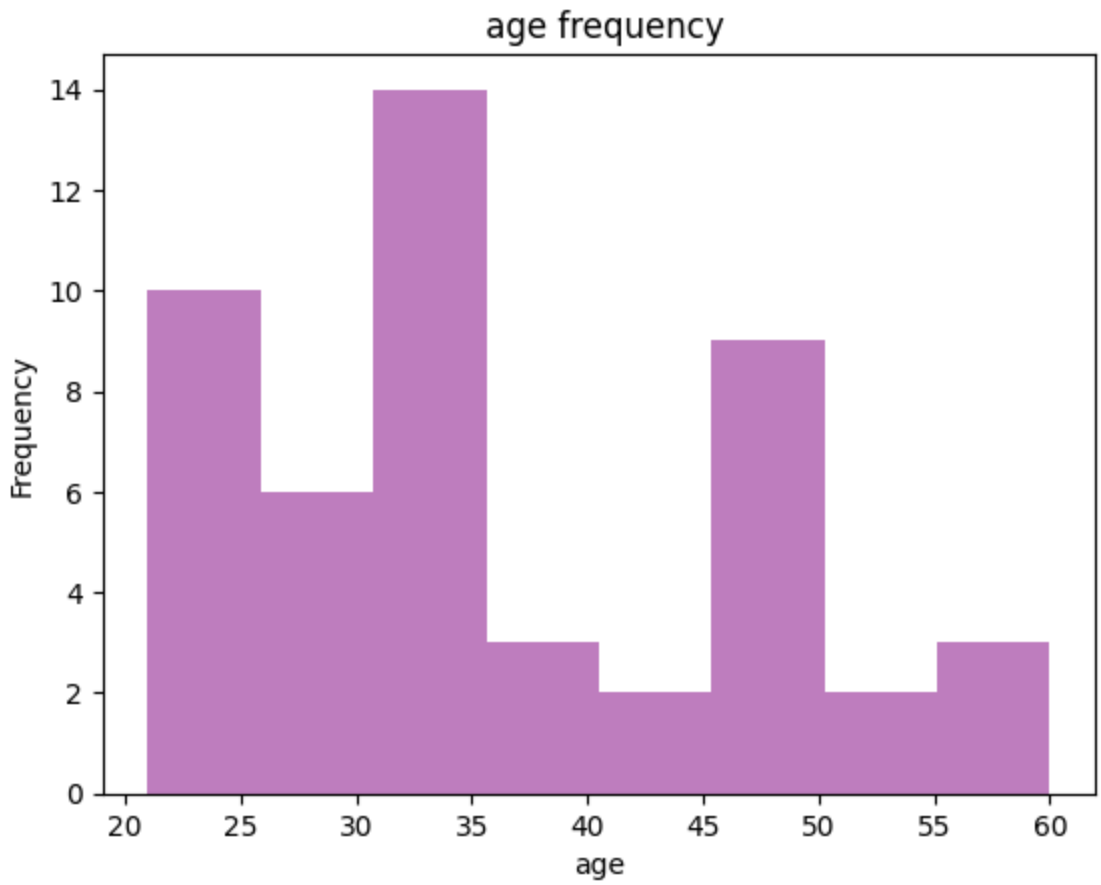
In [ ]:
```
# Compare salary
plt.bar(df['employeeID'], df['salary'], color='blue')
plt.title("employees salary")
plt.xlabel("employeeID")
plt.ylabel("Salary")
plt.show()
```



from this plot we realize that: employee no. 1 has the lowest salary & employee no. 9 has the highest salary

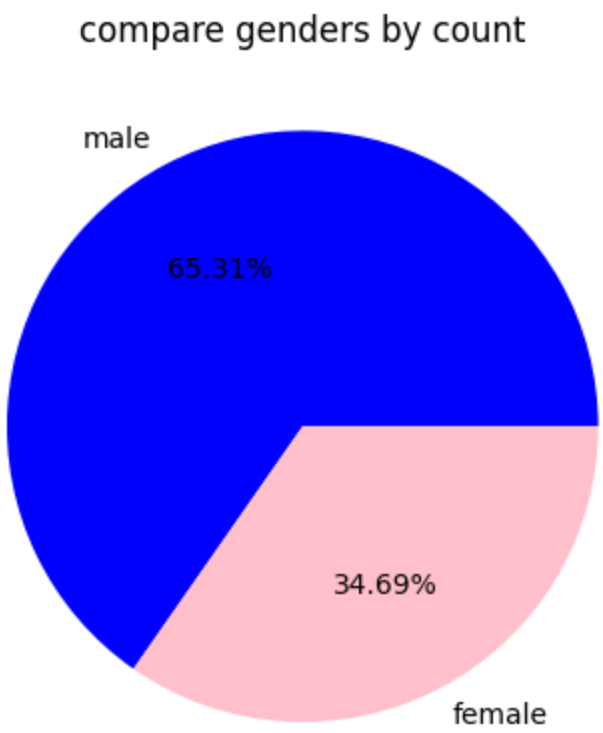
The salaries starts from 2k to 15k

In [ ]:
```
# age frequency
plt.hist(df["age"], bins=8, color = 'purple',alpha=0.5)

plt.title(" age frequency")
plt.xlabel("age")
plt.ylabel("Frequency")
plt.show()
```

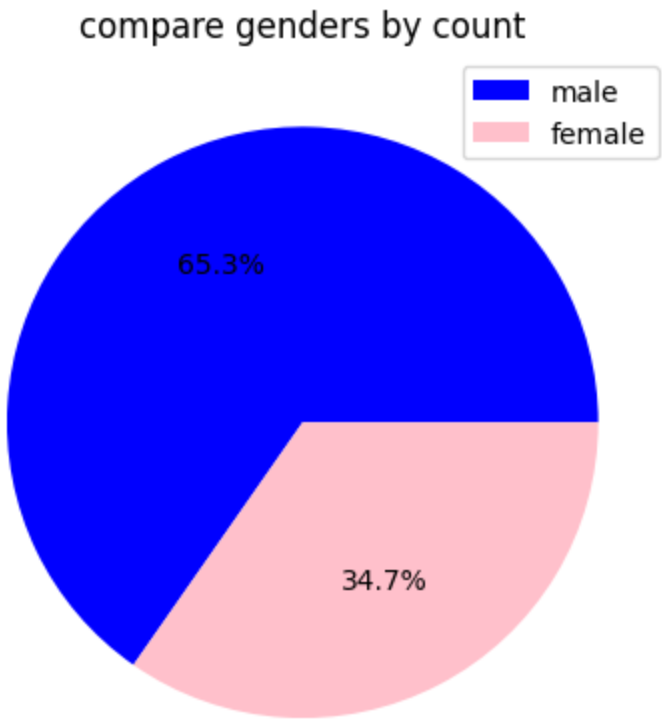


We can see that the age of most employees' is from 30 to 35 years old.

In [ ]:
```
# Compare genders based on employees count
# Group by 'gender' and count the number of employees in each group
gender_counts = df['gender'].value_counts()

# Define colors for the pie chart slices
colors = ['blue', 'pink']

# Create the pie chart
#plt.figure(figsize=(8, 8)) # Set a good figure size
plt.pie(gender_counts, labels=['male','female'], colors=colors, autopct='%.2f%%')

# Add a title to the pie chart
plt.title('compare genders by count')

# Display the plot
plt.show()
```

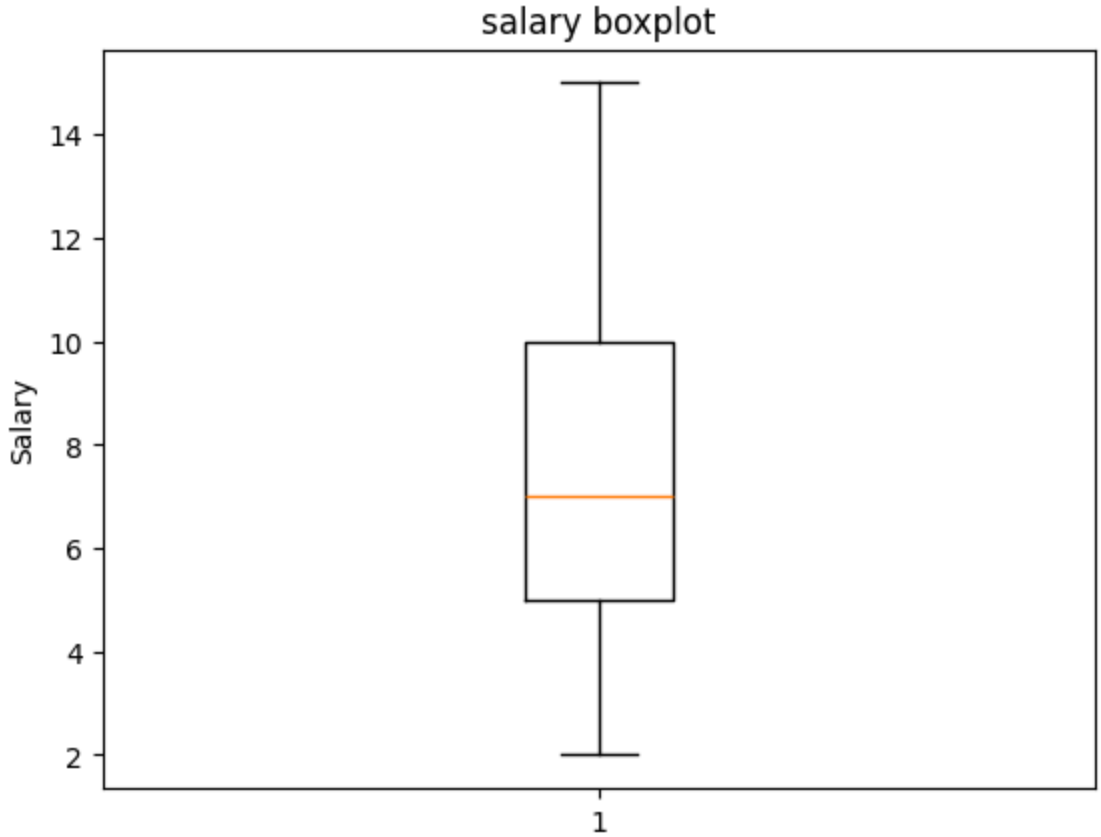


In [ ]:
```
# Add a legend to the pie chart

plt.pie(gender_counts, colors=colors, autopct='%.1f%%')

# Add a title to the pie chart
plt.title('compare genders by count')

# Add a legend
plt.legend(labels=['male','female'], loc="best")
# Display the plot
plt.show()
```



from the pie chart We Realize That Sum of male employees is double the females

In [ ]:
```
# salary boxplot
plt.boxplot(df['salary'])
plt.title("salary boxplot")
plt.ylabel("Salary")
plt.show()
```



from this plot we understand that:

• The minimum salary is 2k and the maximum salary is 15

• Median = 7, q1 = 5 , q3= 10

• There are no outliers

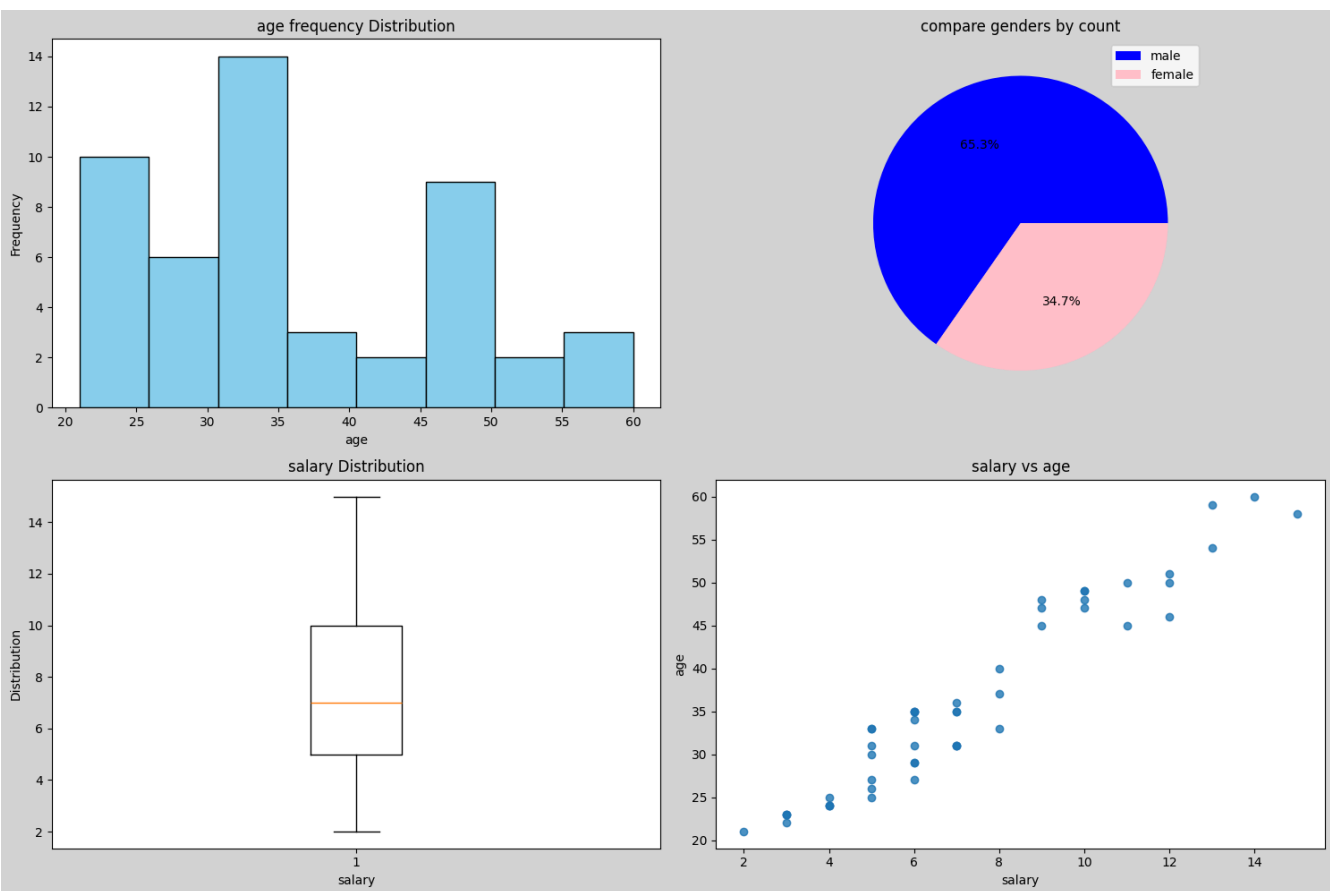
In [ ]:
```
#Let's see if there is a relation between salary & age

#Creating the scatter plot
plt.scatter(df["salary"], df["age"],alpha=0.8)

#Adding the aesthetics
plt.title('salary vs age')
plt.xlabel('salary')
plt.ylabel('age')
```

Out[ ]:
```
Text(0, 0.5, 'age')
```

We can see a positive correlation between salary and age

Combining All Plots in one Dashboard

plt.subplots()

In [ ]:
```
# Create a 2x2 subplot layout
fig, axs = plt.subplots(2, 2, figsize=(15, 10))
```

```python
# Set the bacground color to gray
fig.set_facecolor('lightgrey')

# First figure row 0 and column 0
axs[0, 0].hist(df["age"], color='skyblue', bins=8, edgecolor='black')
axs[0, 0].set_title('age frequency Distribution')
axs[0, 0].set_xlabel('age')
axs[0, 0].set_ylabel('Frequency')

# Second one row 0 and column 1
axs[0, 1].pie(gender_counts, colors=colors, autopct='%.1f%%')
axs[0, 1].set_title('compare genders by count')
axs[0, 1].legend(labels=['male','female'])


# Third one row 1 and column 0
axs[1, 0].boxplot(df["salary"])
axs[1, 0].set_title('salary Distribution')
axs[1, 0].set_xlabel('salary')
axs[1, 0].set_ylabel('Distribution')

# Histogram for Petal Width
axs[1, 1].scatter(df["salary"], df["age"],alpha=0.8)
axs[1, 1].set_title('salary vs age')
axs[1, 1].set_xlabel('salary')
axs[1, 1].set_ylabel('age')


# Adjust layout for better spacing
plt.tight_layout()
plt.show()
```